

Solution to Matrix Chain Multiplication Problem

Question:

Assume that multiplying a matrix G_1 of dimension $p \times q$ with another matrix G_2 of dimension $q \times r$ requires pqr scalar multiplications. Computing the product of n matrices $G_1G_2G_3 \dots G_n$ can be done by parenthesizing in different ways.

Define G_iG_{i+1} as an explicitly computed pair for a given parenthesization if they are directly multiplied.

For example, in the matrix multiplication chain

$$G_1G_2G_3G_4G_5G_6$$

using parenthesization

$$(G_1(G_2G_3))(G_4(G_5G_6))$$

the explicitly computed pairs are:

$$G_2G_3 \quad \text{and} \quad G_5G_6$$

Consider the matrix multiplication chain

$$F_1F_2F_3F_4F_5$$

where matrices are of dimensions:

$$F_1 : 2 \times 25$$

$$F_2 : 25 \times 3$$

$$F_3 : 3 \times 16$$

$$F_4 : 16 \times 1$$

$$F_5 : 1 \times 1000$$

In the parenthesization that minimizes the total number of scalar multiplications, the explicitly computed pair(s) is/are:

(A) F_1F_2 and F_3F_4 only

(B) F_2F_3 only

(C) F_3F_4 only

(D) F_1F_2 and F_4F_5 only

Solution:

The cost of multiplying a matrix of dimension

$$(p \times q)$$

with another matrix of dimension

$$(q \times r)$$

is:

$$pqr$$

scalar multiplications.

The matrix dimensions are:

$$F_1 = 2 \times 25$$

$$F_2 = 25 \times 3$$

$$F_3 = 3 \times 16$$

$$F_4 = 16 \times 1$$

$$F_5 = 1 \times 1000$$

Thus, the dimension array is:

$$[2, 25, 3, 16, 1, 1000]$$

Step 1: Compute Cost of Adjacent Matrix Multiplications**1. Multiplying $F_1 F_2$**

$$(2 \times 25)(25 \times 3)$$

Cost:

$$2 \times 25 \times 3 = 150$$

Resulting matrix dimension:

$$2 \times 3$$

2. Multiplying F_2F_3

$$(25 \times 3)(3 \times 16)$$

Cost:

$$25 \times 3 \times 16 = 1200$$

Resulting matrix dimension:

$$25 \times 16$$

3. Multiplying F_3F_4

$$(3 \times 16)(16 \times 1)$$

Cost:

$$3 \times 16 \times 1 = 48$$

Resulting matrix dimension:

$$3 \times 1$$

4. Multiplying F_4F_5

$$(16 \times 1)(1 \times 1000)$$

Cost:

$$16 \times 1 \times 1000 = 16000$$

Resulting matrix dimension:

$$16 \times 1000$$

Step 2: Choose the Minimum Cost Multiplication

Among all adjacent pairs:

$$F_1F_2 = 150$$

$$F_2F_3 = 1200$$

$$F_3F_4 = 48$$

$$F_4F_5 = 16000$$

The minimum cost is:

$$F_3 F_4 = 48$$

Hence, the optimal parenthesization should compute:

$$(F_3 F_4)$$

first.

Step 3: Continue Remaining Multiplications

After computing:

$$(F_3 F_4)$$

the resulting matrix has dimension:

$$3 \times 1$$

Now the chain becomes:

$$F_1(2 \times 25), \quad F_2(25 \times 3), \quad (F_3 F_4)(3 \times 1), \quad F_5(1 \times 1000)$$

Multiply $F_2(F_3 F_4)$

$$(25 \times 3)(3 \times 1)$$

Cost:

$$25 \times 3 \times 1 = 75$$

Resulting matrix:

$$25 \times 1$$

Multiply F_1 with the above result

$$(2 \times 25)(25 \times 1)$$

Cost:

$$2 \times 25 \times 1 = 50$$

Resulting matrix:

$$2 \times 1$$

Multiply with F_5

$$(2 \times 1)(1 \times 1000)$$

Cost:

$$2 \times 1 \times 1000 = 2000$$

Step 4: Compute Total Cost

Total number of scalar multiplications:

$$48 + 75 + 50 + 2000$$

$$= 2173$$

This is the minimum possible cost.

Step 5: Determine Explicitly Computed Pairs

An explicitly computed pair means two original adjacent matrices that are directly multiplied.

In the optimal parenthesization, only:

$$F_3F_4$$

is directly multiplied.

Therefore, the explicitly computed pair is:

$$\boxed{F_3F_4 \text{ only}}$$

Final Answer:

$$\boxed{(C) \ F_3F_4 \text{ only}}$$